Bipin Shrestha

Monroe, LA | bipinshrestha830@gmail.com | +1 318-690-8483 | GitHub

SUMMARY

Full Stack Developer with a Bachelor of Science in Computer Science and strong experience developing scalable web applications using Java, Spring Boot, React, JavaScript, SQL, and cloud technologies. Experienced in designing responsive user interfaces, developing RESTful APIs, managing relational databases, and deploying applications using AWS and Docker. Passionate about building reliable, data-driven applications, solving complex technical problems, and collaborating with cross-functional teams to deliver high-quality software solutions. Proficient in Java with strong algorithms and data structures skills, experienced in full-stack web development and cloud/DevOps projects, AWS Certified Solutions Architect and Developer Associate, eager to contribute innovative solutions and grow in a dynamic tech environment.

EDUCATION

University of Louisiana at Monroe — Monroe, LA

Bachelor of Science in Computer Science

Data Structures and Algorithms, Object-Oriented Programming, Operating System, Computer Networks, Database Management

Technical Skill

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5 / CSS3, Tailwind CSS, Chart.js, Responsive UI, Leaflet.js, GIS basics, Open Layers, NOAA Weather API

Backend & APIs: Java, C++, Python, C, Go, Spring Boot, Spring AI, REST APIs, Git, JWT Authentication & Role-Based Access, AOP, Microservices, Kafka, JUnit Testing, Selenium

Databases & Data: MySQL, SQL, PostgreSQL, MongoDB, Hibernate/JPA ORM, Data Modeling, Caching

AI & Automation: OpenRouter API, Generative AI, Spring AI, n8n, MCP

DevOps & Cloud / Project Management: Docker, Kubernetes, RabbitMQ, AWS (API Gateway, Elastic Beanstalk, SNS, SQS, DynamoDB), CI/CD , Linux, GitHub, Jira, Scrum, Kanban, Jenkins

PROJECTS

Grade Forge – Automated Programming Assignment Grading Platform *Repository*

A scalable platform for managing programming assignments where instructors can create test suites and automatically evaluate student code submissions using containerized execution.

- Led the backend development of a **team-based project**, designing core system architecture
- Built a secure platform where **Faculty create assignments and test suites**, and students submit code that is compiled and executed inside **isolated Docker containers** for automated grading.
- Implemented a **test case execution engine** that evaluates submissions against multiple test cases, storing compile results, runtime output, execution time, and pass/fail status for each submission.
- Designed **role-based access control** (Faculty, TA, Student) using Spring Security to manage assignment creation, grading workflows, and submission tracking.
- Designed normalized **PostgreSQL database** schemas supporting thousands of assignment submissions and grading records.
- Optimized SQL queries to improve application performance and reduce database response times.
- Deployed the system on AWS EC2 with **AWS RDS (PostgreSQL)** for database management and **AWS S3** for storing submission files and artifacts.
- Designed the system to support **future enhancements** such as distributed code runners, asynchronous grading queues, plagiarism detection, and grading analytics dashboards.
- Applied strong communication and critical thinking skills to coordinate with team members, solve technical challenges, and successfully deliver the project.

Tech Stack: Java, Spring Boot, React, Tailwind CSS, Docker, PostgreSQL, AWS EC2, AWS RDS, AWS S3, Lambda Functions, AWS CodeBuild, AWS IAM, Spring Security, REST APIs, GitHub, Hibernate

Hotel Booking Application *Repository*

A full-stack hotel booking system for privately-owned or large hotels, built with Spring Boot, Next.js, and AWS.

- Implemented secure JWT-based authentication with role-based access for users, supporting sign-up, login, bookings, payments, and profile management.
- Developed admin dashboard for room management (CRUD) and booking approvals, integrated with asynchronous email notifications and Stripe payments.
- Deployed backend on AWS with CI/CD, reducing **deployment time from 30 mins to <5 mins**.

Tech Stack: *Java, Spring Boot, Spring, Next.js, AWS, MySQL, Tailwind CSS, JavaScript*

Research & Collaboration

Emerging Scholar Program Research *Spring 2023 – Fall 2023*

Academic Research -University of Louisiana at Monroe

- VR in Education – analyzed teacher perceptions of virtual reality for enhanced student engagement and experiential learning
- AI in Business Communication – Studied impacts of AI, Robotics, and ICT on modern business operations, workforce Management, and global economic trends.

Google Developer Student Club (GDSC) *Fall 2024 - Present*

Active Community Member – Coding competition, hackathons, and mentoring fellow developers.

Certifications

AWS Certified Developer – Associate (Amazon Web Services, 2026)
AWS Certified Solutions Architect – Associate (Amazon Web Services, 2025)
AWS AI Practitioner (Amazon Web Services, 2025)
AWS Certified Cloud Practitioner (Amazon Web Services, 2024)
Professional Certificate in DevOps (Udemy, 2025)
Linux Command Line: From Zero to Hero (Udemy, 2025)
Problem Solving (Java) – HackerRank (2024)